

Appendix B: Simple Connectivity and Jordan Curve Theorem

Jordan was one of the precursors of the theory of functions of a real variable. He introduced in this part of analysis the capital notion of functions of bounded variation. Not less celebrated is his study of curves, universally called Jordan curves, which curves separate the plane in two distinct regions. We also owe him important propositions regarding the measure of sets that have led the way to numerous modern researches.

E. Picard, 1922

The notion of simple connectivity is at the source of many basic and fundamental results in complex analysis. To clarify the meaning of this important concept, we have gathered in this appendix some further insights into the properties of simply connected sets. Closely tied to the idea of simple connectivity is the notion of the “interior” of a simple closed curve. The theorem of Jordan states that this interior is well-defined and is simply connected. We prove here the special case of this theorem for curves which are piecewise-smooth.

Recall the definition in Chapter 3, according to which a region Ω is simply connected if any two curves in Ω with the same end-points are homotopic. From this definition we deduced an important version of Cauchy’s theorem which states that if Ω is simply connected and $\gamma \subset \Omega$ is any closed curve, then

$$(1) \quad \int_{\gamma} f(\zeta) d\zeta = 0$$

whenever f is holomorphic in Ω . Here, we shall prove that a converse also holds, therefore:

- (I) A region Ω is simply connected if and only if it is holomorphically simply connected; that is, whenever $\gamma \subset \Omega$ is closed and f holomorphic in Ω then (1) holds.

Besides this fundamental equivalence, which is analytic in nature, there

are also topological conditions that can be used to describe simple connectivity. In fact, the definition in terms of homotopies suggests that a simply connected set has no “holes.” In other words, one cannot find a closed curve in Ω that loops around points that do not belong to Ω . In the first part of this appendix we shall also turn these intuitive statements into tangible theorems:

- (II) We show that a bounded region Ω is simply connected if and only if its complement is connected.
- (III) We define the winding number of a curve around a point, and prove that Ω is simply connected if and only if no curve in Ω winds around points in the complement of Ω .

In the second part of this appendix we return to the problem of curves and their interior. The main question is the following: given a closed curve Γ that does not intersect itself (it is simple), can we make sense of the “region enclosed by Γ ”? In other words, what is the “interior” of Γ ? Naturally, we may expect the interior to be open, bounded, simply connected, and have Γ as its boundary. To solve this problem, at least when the curve is piecewise-smooth, we prove a theorem that guarantees the existence of a unique set which satisfies all the desired properties. This is a special case of the Jordan curve theorem, which is valid in the general case when the simple curve is assumed to be merely continuous. In particular, our result leads to a generalization of Cauchy’s theorem in Chapter 2 which we formulated for toy contours.

We continue to follow the convention set in Chapter 1 by using the term “curve” synonymously with “piecewise-smooth curve,” unless stated otherwise.

1 Equivalent formulations of simple connectivity

We first dispose of (I).

Theorem 1.1 *A region Ω is holomorphically simply connected if and only if Ω is simply connected.*

Proof. One direction is simply the version of Cauchy’s theorem in Corollary 5.3, Chapter 3. Conversely, suppose that Ω is holomorphically simply connected. If $\Omega = \mathbb{C}$, then it is clearly simply connected. If Ω is not all of $\mathbb{C}$, recall that the Riemann mapping theorem still applies (see the remark following its proof in Chapter 8), hence Ω is conformally equivalent to the unit disc. Since the unit disc is simply connected, the same must be true of Ω .

Next, we turn to (II) and (III), which, as we mentioned, are both precise formulations of the fact that a simply connected region cannot have “holes.”

Theorem 1.2 *If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if the complement of Ω is connected.*

Note that we assume that Ω is bounded. If this is not the case, then the theorem as stated does not hold, for example an infinite strip is simply connected yet its complement consists of two components. However, if the complement is taken with respect to the extended complex plane, that is, the Riemann sphere, then the conclusion of the theorem holds regardless of whether Ω is bounded or not.

Proof. We begin with the proof that if Ω^c is connected, then Ω is simply connected. This will be achieved by showing that Ω is holomorphically simply connected. Therefore, let γ be a closed curve in Ω and f a holomorphic function on Ω . Since Ω is bounded, the set¹

$$K = \{z \in \Omega : d(z, \Omega^c) \geq \epsilon\}$$

is compact, and for sufficiently small ϵ , the set K contains γ . In an attempt to apply Runge’s theorem (Theorem 5.7 in Chapter 2), we must first show that the complement K^c of K is connected.

If this is not the case, then K^c is the disjoint union of two non-empty open sets, say $K^c = \mathcal{O}_1 \cup \mathcal{O}_2$. Let

$$F_1 = \mathcal{O}_1 \cap \Omega^c \quad \text{and} \quad F_2 = \mathcal{O}_2 \cap \Omega^c.$$

Clearly, $\Omega^c = F_1 \cup F_2$, so if we can show that F_1 and F_2 are disjoint, closed, and non-empty, then we will conclude that Ω^c is not connected, thus contradicting the hypothesis in the theorem. Since $\mathcal{O}_1$ and $\mathcal{O}_2$ are disjoint, so are F_1 and F_2 . To see why F_1 is closed, suppose $\{z_n\}$ is a sequence of points in F_1 that converges to z . Since Ω^c is closed we must have $z \in \Omega^c$, and since Ω^c is at a finite distance from K , we deduce that $z \in \mathcal{O}_1 \cup \mathcal{O}_2$. Now we observe that we cannot have $z \in \mathcal{O}_2$, for otherwise we would have $z_n \in \mathcal{O}_2$ for sufficiently large n because $\mathcal{O}_2$ is open, and this contradicts the fact that $z_n \in F_1$ and $\mathcal{O}_1 \cap \mathcal{O}_2 = \emptyset$. Hence $z \in \mathcal{O}_1$ and F_1 is closed, as desired. Finally, we claim that F_1 is non-empty. If otherwise, $\mathcal{O}_1$ is contained in Ω . Select any point $w \in \mathcal{O}_1$, and since $w \notin K$, there exists $z \in \Omega^c$ with $|w - z| < \epsilon$, and the entire line segment from w to z belongs to K^c . Since $z \in \mathcal{O}_2$ (because $\mathcal{O}_1 \subset \Omega$), some point

¹Here, $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$ denotes the distance from z to Ω^c .

on the line segment $[z, w]$ must belong to neither $\mathcal{O}_1$ nor $\mathcal{O}_2$, and this is a contradiction. More precisely, if we set

$$t^* = \sup\{0 \leq t \leq 1 : (1-t)z + tw \in \mathcal{O}_2\},$$

then $0 < t^* < 1$, and the point $(1-t^*)z + t^*w$, which is not in K , cannot belong to either $\mathcal{O}_1$ or $\mathcal{O}_2$ since these sets are open. Similar arguments imply the same conclusions for F_2 , and we have reached the desired contradiction. Thus K^c is connected.

Therefore, Runge's theorem guarantees that f can be approximated uniformly on K , and hence on γ , by polynomials. However, $\int_{\gamma} P(z) dz = 0$ whenever P is a polynomial, so in the limit we conclude that $\int_{\gamma} f(z) dz = 0$, as desired.

The converse result, that Ω^c is connected whenever Ω is bounded and simply connected, will follow from the notion of winding numbers, which we discuss next.

Winding numbers

If γ is a closed curve in $\mathbb{C}$ and z a point not lying on γ , then we may calculate the number of times the curve γ winds around z by looking at the change of argument of the quantity $\zeta - z$ as ζ travels on γ . Every time γ loops around z , the quantity $(1/2\pi) \arg(\zeta - z)$ increases (or decreases) by 1. If we recall that $\log w = \log |w| + i \arg w$, and denote the beginning and ending points of γ by ζ_1 and ζ_2 , then we may guess that the quantity

$$\frac{1}{2\pi i} [\log(\zeta_1 - z) - \log(\zeta_2 - z)], \quad \text{which "equals"} \quad \frac{1}{2\pi i} \int_{\gamma} \frac{d\zeta}{\zeta - z},$$

computes precisely the number of times γ loops around ζ .

These considerations lead to the following precise definition: the **winding number** of a closed curve γ around a point $z \notin \gamma$ is

$$W_{\gamma}(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{d\zeta}{\zeta - z}.$$

Sometimes, $W_{\gamma}(z)$ is also called the index of z with respect to γ .

For example, if $\gamma(t) = e^{ikt}$, $0 \leq t \leq 2\pi$, is the unit circle traversed k times in the positive direction (with $k \in \mathbb{N}$), then $W_{\gamma}(0) = k$. In fact, one has

$$W_{\gamma}(z) = \begin{cases} k & \text{if } |z| < 1, \\ 0 & \text{if } |z| > 1. \end{cases}$$

Similarly, if $\gamma(t) = e^{-ikt}$, $0 \leq t \leq 2\pi$, is the unit circle traversed k times in the negative direction, then we find that $W_\gamma(z) = -k$ in the interior of the disc, and $W_\gamma(z) = 0$ in its exterior.

Note that, if γ denotes a positively oriented toy contour, then

$$W_\gamma(z) = \begin{cases} 1 & \text{if } z \in \text{interior of } \gamma, \\ 0 & \text{if } z \in \text{exterior of } \gamma. \end{cases}$$

In general we have the following natural facts about winding numbers.

Lemma 1.3 *Let γ be a closed curve in $\mathbb{C}$.*

- (i) *If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.*
- (ii) *If z and w belong to the same open connected component in the complement of γ , then $W_\gamma(z) = W_\gamma(w)$.*
- (iii) *If z belongs to the unbounded connected component in the complement of γ , then $W_\gamma(z) = 0$.*

Proof. To see why (i) is true, suppose that $\gamma : [0, 1] \rightarrow \mathbb{C}$ is a parametrization for the curve, and let

$$G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z} ds.$$

Then G is continuous and, except possibly at finitely many points, it is differentiable with $G'(t) = \gamma'(t)/(\gamma(t) - z)$. This implies that, except possibly at finitely many points, the derivative of the continuous function $H(t) = (\gamma(t) - z)e^{-G(t)}$ is zero, and hence H must be constant. Putting $t = 0$ and recalling that γ is closed, so that $\gamma(0) = \gamma(1)$, we find

$$1 = e^{G(0)} = c(\gamma(0) - z) = c(\gamma(1) - z) = e^{G(1)}.$$

Therefore, $G(1)$ is an integral multiple of $2\pi i$, as desired.

For (ii), we simply note that $W_\gamma(z)$ is a continuous function of $z \notin \gamma$ that is integer-valued, so it must be constant in any open connected component in the complement of γ .

Finally, one observes that $\lim_{|z| \rightarrow \infty} W_\gamma(z) = 0$, and, combined with (ii), this establishes (iii).

We now show that the notion of a bounded simply connected set Ω may be understood in the following sense: no curve in Ω winds around points in Ω^c .

Theorem 1.4 *A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for any closed curve γ in Ω and any point z not in Ω .*

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω , and Cauchy's theorem gives $W_\gamma(z) = 0$.

For the converse, it suffices to prove that the complement of Ω is connected (Theorem 1.2). We argue by contradiction, and construct an explicit closed curve γ in Ω and find a point w so that $W_\gamma(w) \neq 0$.

If we suppose that Ω^c is not connected, then we may write $\Omega^c = F_1 \cup F_2$ where F_1, F_2 are disjoint, closed, and non-empty. Only one of these sets can be unbounded, so that we may assume that F_1 is bounded, thus compact. The curve γ will be constructed as part of the boundary of an appropriate union of squares.

Lemma 1.5 *Let w be any point in F_1 . Under the above assumptions, there exists a finite collection of closed squares $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ that belong to a uniform grid $\mathcal{G}$ of the plane, and are such that:*

- (i) w belongs to the interior of Q_1 .
- (ii) The interiors of Q_j and Q_k are disjoint when $j \neq k$.
- (iii) F_1 is contained in the interior of $\bigcup_{j=1}^n Q_j$.
- (iv) $\bigcup_{j=1}^n Q_j$ is disjoint from F_2 .
- (v) The boundary of $\bigcup_{j=1}^n Q_j$ lies entirely in Ω , and consists of a finite union of disjoint simple closed polygonal curves.

Assuming this lemma for now, we may easily finish the proof of the theorem. The boundary ∂Q_j of each square is equipped with the positive orientation. Since $w \in Q_1$, and $w \notin Q_j$ for all $j > 1$, we have

$$(2) \quad \sum_{j=1}^n \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1.$$

If $\gamma_1, \dots, \gamma_M$ denotes the polygonal curves in (v) of the lemma, then, the cancellations arising from integrating over the same side but in opposite directions in (2) yield

$$\sum_{j=1}^n \frac{1}{2\pi i} \int_{\gamma_j} \frac{d\zeta}{\zeta - w} = 1,$$

and hence $W_{\gamma_{j_0}}(w) \neq 0$ for some j_0 . The closed curve γ_{j_0} lies entirely in Ω , and this gives the desired contradiction.

Proof of the lemma. Since F_2 is closed, the sets F_1 and F_2 are at a finite non-zero distance d from one another. Now consider a uniform grid $\mathcal{G}_0$ of the plane consisting of closed squares of side length which is much smaller than d , say $< d/100$, and such that w lies at the center of a closed square R_1 in this grid. Let $\mathcal{R} = \{R_1, \dots, R_m\}$ denote the finite collection of all closed squares in the grid that intersect F_1 . Then, the collection $\mathcal{R}$ satisfies properties (i) through (iv) of the lemma. To guarantee (v), we argue as follows.

The boundary of each square in $\mathcal{R}$ is given the positive (counterclockwise) orientation. The boundary of $\bigcup_{j=1}^m R_j$ is then equal to the union of all boundary sides, that is, those sides that do not belong to two adjacent squares in the collection $\mathcal{R}$. Similarly, the boundary vertices are the end-points of all boundary sides. A boundary vertex is said to be “bad,” if it is the end-point of more than two boundary sides. (See point P on Figure 1.)

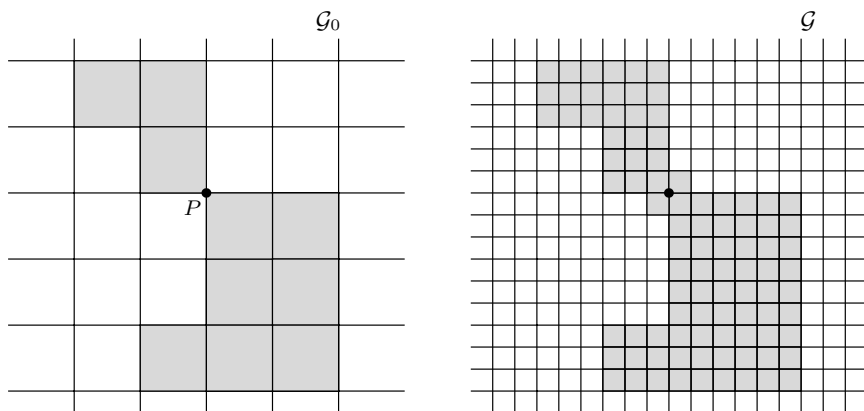


Figure 1. Eliminating bad boundary vertices

To eliminate the bad boundary vertices, we refine the grid $\mathcal{G}_0$ and possibly add some squares. More precisely, consider the grid $\mathcal{G}$ obtained as a refinement of the original grid, by dissecting all squares of $\mathcal{G}_0$ into nine equal subsquares. Then, let $Q_1, \dots, Q_p$ denote all the squares in the grid $\mathcal{G}$ that are subsquares of squares in the collection $\mathcal{R}$ (so in particular, $p = 9n$), and where Q_1 is chosen so that $w \in Q_1$. Then, we may add finitely many squares from $\mathcal{G}$ near each bad boundary vertex, so that the resulting family $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ has no bad boundary vertices. (See Figure 1.)

Clearly, $\mathcal{Q}$ still satisfies (i) through (iv), and we claim this collection also satisfies (v). Indeed, let $[a_1, a_2]$ denote any boundary side of $\bigcup_{j=1}^n Q_j$ with its orientation from a_1 to a_2 . By considering the three different possibilities, one sees that a_2 is the beginning point of another boundary side $[a_2, a_3]$. Continuing in this fashion, we obtain a sequence of boundary sides $[a_1, a_2], [a_2, a_3], \dots, [a_n, a_{n+1}], \dots$. Since there are only finitely many sides, we must have $a_n = a_m$ for some n and some $m > n$. We may choose the smallest m so that $a_n = a_m$, say $m = m'$. Then, we note that if $n > 1$, then $a_{m'}$ is an end-point of at least three boundary sides, namely $[a_{n-1}, a_n]$, $[a_n, a_{n+1}]$, and $[a_{m'-1}, a_{m'}]$, hence $a_{m'}$ is a bad boundary vertex. Since we arranged that $\mathcal{Q}$ had no such boundary vertices, we conclude that $n = 1$, and hence the polygon formed by $a_1, \dots, a_{m'}$ is closed and simple. We may repeat this process and find that $\mathcal{Q}$ satisfies property (v), and the proof of Lemma 1.5 is complete.

Finally, we are now able to finish the proof of Theorem 1.2, namely, if Ω is bounded and simply connected we can conclude that Ω^c is connected. To see this, note that if Ω^c is not connected, then we have constructed a curve $\gamma \subset \Omega$ and found a point $w \notin \Omega$ so that $W_\gamma(w) \neq 0$, thus contradicting the fact that Ω is simply connected.

2 The Jordan curve theorem

Although we emphasize in the statement of the theorems which follow that the curves are piecewise-smooth, we note that the proofs involve the use of curves that may only be continuous, (the curves Γ_ϵ below).

The two main results in this section are the following.

Theorem 2.1 *Let Γ be curve in the plane that is simple and piecewise-smooth. Then, the complement of Γ is an open connected set whose boundary is precisely Γ .*

Theorem 2.2 *Let Γ be a curve in the plane which is simple, closed, and piecewise-smooth. Then, the complement of Γ consists of two disjoint connected open sets. Precisely one of these regions is bounded and simply connected; it is called the **interior** of Γ and denoted by Ω . The other component is unbounded, called the **exterior** of Γ , and denoted by $\mathcal{U}$.*

Moreover, with the appropriate orientation for Γ , we have

$$W_\Gamma(z) = \begin{cases} 1 & \text{if } z \in \Omega, \\ 0 & \text{if } z \in \mathcal{U}. \end{cases}$$